



HAZOP Leadership training for GC

Chapter 2 - Overview PHA techniques

Technical Safety Service Division (Q-TS-TS), 1 - 3 November 2021

Activity-3 : List PHA techniques (Menti-3)

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List PHA techniques that you have been used

<https://www.menti.com/1g4hx55ryn>



- ☐ Process hazard analysis (PHA) techniques overview
- ☐ Check list
- ☐ What-if
- ☐ FMEA
- ☐ Safety Project Life Cycle Effective risk management through smarter project planning

Ways to Identify Risk

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- ☐ Hazard identification is the first step in a systematic process which aims to improve the management of risk.
- ☐ A number of techniques can be used in the hazard identification process.
- ☐ Each technique has advantages and disadvantages, and care should be taken in choosing the most appropriate one for each complexity.

What is your appropriate technique?



Hazard Evaluation Techniques

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1960 – 2001

Safety Review

Walk Through
Inspection



Check Lists

Historical Lists
Yes / No

1965 – 2001

Relative Ranking

ICI Mond index
Dow FEI



1972 - 2001

What-if

Brainstorming



1974 – 2001

HAZOP

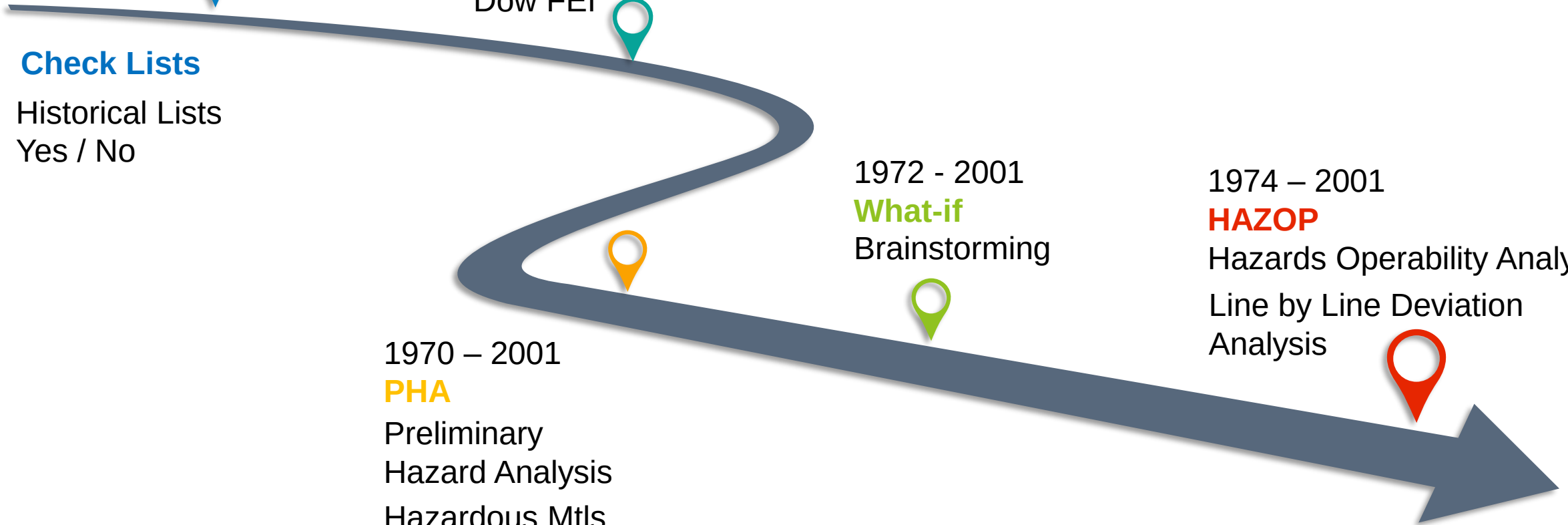
Hazards Operability Analysis
Line by Line Deviation
Analysis



1970 – 2001

PHA

Preliminary
Hazard Analysis
Hazardous Mtls
Hazardous Opns



Hazard Evaluation Techniques

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Several of these identification methods are incorporated into process Hazards Assessment

☐ Checklist

Probably the simplest method using a tabulated series of questions

☐ “What-if” study

Carried out using a brainstorming technique typically starting from hazards known to the group leading on to other potential scenarios Cited in OSHA 1910.119

☐ HAZOP Study (Explain detail in Chapter 3 & 4)

A group review using structured questioning to focus on deviations from design intent which may create hazard or operability problems.

☐ Failure Mode and Effects Analysis (FMEA)

A component-by-component assessment of the ways of failure of each item of equipment in a system and the effects on system operate that result

☐ Fault-Tree Analysis

Fault tree analysis (FTA) is a graphical tool to explore the causes of system level failures. It uses boolean logic to combine a series of lower level events and it is basically a top-down approach to identify the component level failures (basic event) that cause the system level failure (top event) to occur.

IEAT PSM

Allowable PHA technique

- What-if
- Checklist
- What-if/Checklist
- Hazard and Operability Study (HAZOP)
- Failure Mode and Effects Analysis (FMEA)
- Fault Tree Analysis
- Other equivalent Techniques



Checklist

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☐ Specific

- Usually developed from broad-based experience or consensus (or industry) standards or codes
- Easily used by an individual analyst or reviewer to compare system design against standards
- Little, if any, creativity on the part of the reviewer(s)



☐ Open

- Relies heavily on the experience of the [reviewer](#)(s)
- Appropriate for use by a multi-disciplined team with broad-based experience
- Application requires creativity on the part of reviewers

Checklist (Prelim-SHE)

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[illegible]

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3C Recommendations

1. To review log requirements
2. To review / update the safety data sheet for the chemicals in the systems
3. To review the relevant standards / codes
4. To review / update the process condition limits e.g. in Operating Windows
5. To review / update the operation, maintenance procedure
6. To review blast / heat / toxic / risk contour
7. To review plot plan, safety distance
8. To review the design can be easily access for operation / maintenance
9. To review PPE, safety equipment, eye washer, safety shower, wind sock
10. To review fire prevention, fire fighting systems, pre-incident plan
11. To review equipment label / pipe label / warning sign
12. To review relief / blowdown capacity
13. To review pressure / vacuum systems of container, vessel, tank
14. To review piping / equipment material

15. To review structure load
16. To review fire proofing
17. To review electrical area classification
18. To review lightning, bonding, grounding systems
19. To review vent and flare system
20. To review power supply, utilities system
21. To review DCS & safeguarding systems
22. To review rainwater, wastewater drainage system
23. To review laydown area
24. To review food / drinking / sanitary
25. To review solid waste / garbage / disposal system
26. To review soil / underwater contamination
27. To review equipment guard / area barrier / fence
28. To review security system

29. To review emergency, means of escape, first aid
30. To review feed / product qualities
31. To update PFD, P&ID, drawings, datasheet
32. To review manpower, training
33. To review employee health check program
34. To review energy index, energy use
35. To review VOC, GHG
36. To review EIA / HIA
37. To review customer requirement
38. To review impact to communities
39. To review impact to company reputation
40. To review storage condition and area in warehouse

No.	3D Proposed Actions refer to Part 3C and/or Additional if any	Responsible person	Due Date		Approve by	Responsible person closed out in ATS	Approver approve (Closed date)
			Estimate date	Within e-MOC Part			
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
14							
15							

Part 4 The team reviews if further PHA (Process Hazard Analysis) is required or not.

4A PHA Questions		4B Conclusion	
Y	N		
☐	☐	☐	"Yes", Further PHA (Process Hazard Analysis) is required.
☐	☐	☐	"No", Further PHA (Process Hazard Analysis) is NOT required.
1 Is the change e.g. multiple piping circuits, multiple P&IDs, large number of equipment items? 2 Could the change take the process outside operating windows (limits of normal operation)? 3 Does the change introduce significant change in the inventory of toxic or flammable materials? 4 Does the change reorder or alter the processing sequence? 5 Does the change significantly impact on the energy balance or mass balance? 6 Does the change alter, affect and/or bypass a safety device or a critical control system component? 7 Does the change need a significant training for staff (e.g. normal, emergency operation)? 8 Does the change introduce or affect a significant source of potential chemical, mechanical, thermal or electrical energy?		Conclusion _____ _____ _____ _____ _____ _____	

Note : Further PHA should be performed if any "Yes" in Question 1-3 or more than 2 "Yes" in Question 4-8

Part 5 Evaluate Hazard Severity Rating of the change if further PHA is **NOT** required.

If **NO** further PHA is required, identify cause and consequence *after the change have been implemented (the worst case)* :

Cause:

Consequence: _____
 Preventive and Corrective Measures: _____

Consequence Rating:

Likelihood Rating:

Severity Rating:

If severity is 5 (Extreme), plant change must be stop.

If severity is 4 (High), plant change must be stop.

If severity is 3 (Medium), action must be proposed.

Likelihood	Consequence	Hazard Severity Rating					
		Low	Medium	High	Extreme	Extreme	Extreme
Frequent (10 ⁻²) Frequent (10 ⁻³) Frequent (10 ⁻⁴) Frequent (10 ⁻⁵)	People (Safety, Health, Welfare) Environment Economic (Low Loss) Small Loss	Low Low Low Low Very Low Very Low	Medium Medium Low Low Very Low Very Low	High High High High High High	Extreme Extreme Extreme Extreme Extreme Extreme	Extreme Extreme Extreme Extreme Extreme Extreme	Extreme Extreme Extreme Extreme Extreme Extreme

Table: Hazard Index (for PART 2B)

Score	Inventory (ton or t / h)	Temperature (Degree C)	Pressure (Barg)	Flash Point (Degree C)	Explosiveness (UEL-LEL, %vol)	Toxicity (TLV, ppm)
0	0-1 t, t / h	0-70	0-5	Non Flammable	Non explosive	> 10000
1	1-10 t, t / h	<0, >70-150	<0, >5-25	> 55 DegC	0-20	>1000-10000
2	10-50 t, t / h	>150-300	> 25-50	> 21-55 DegC	>20-45	>100-1000
3	50-200 t, t / h	>300-600	>50-200	0-21 DegC	>45-70	>10-100
4	200-500 t, t / h	>600	>200-1000	<0 DegC	>70-100	>1-10
5	500-1000 t, t / h		>1000			
6	>1000 t, t / h					<0.1

- ❑ A review team formulates a series of questions that must be answered by the team members or by other subject matter experts.
- ❑ The team then suggests corrective actions when the questions/answers reveal unacceptable risks.

It is an extremely flexible and easily used approach!

- ❑ Example
 - "What if" a specific component fails in a specific condition
 - "What if" a specific process parameter (flow, level, temperature) is abnormal?
 - "What if" a specific operator action or maintenance action is performed incorrectly or not performed at all?
 - "What if" a specific external event or condition occurs?

However, it also require more experience of leader for facilitate and prepare the question list

What – if (Example)

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Session: (2) 21/10/03

Revision: O

Node: (1) BGP Utilities

Intention: (1) Liquid Fuel System -- unload diesel from trucks into Diesel Tank

Quantity unloaded 20,000 litres, at ambient temp. Pump discharge around 5 barg.

Drawings: BCD3-695-42-PID-1-01 1/0121013-01,
BCD3-695-42-PID-1-012-01

WHAT IF...	HAZARD - Issue of Concern	CONSEQUENCES	SAFEGUARDS	S	L	R	REF#	RECOMMENDATIONS	BY
1.Static Electricity During unloading	Ignition source, spark	Build-up of static electricity during unloading creates spark which ignites fuel causing fire	Operator Training & Procedures (OTP) Firefighting equipment (portable and fixed) Emergency Response Procedures-including mutual aid response	2	2	2	1.1.1	Ensure all tanker unloading stations are equipped with adequate grounding/ earthing facilities. Consider whether an active system is warranted.	CSC
							1.1.2	Tanker unloading procedure needs to be prepared and personnel properly trained to ensure all aspects of unloading are undertaken safely	PCI
2.695-P-104 seal fails	Diesel fuel-leak/spill	Leak from seal spill to ground, causing potential environmental spill, with possible localized pool fire	PM program for Pump OTP - attended Operation High flash point & control of ignition sources (HAC) Pump located in concrete diked area, isolated from sewer	3	3	4			

What-if/Checklist (Example)

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Part 1 Piping
Part 2 Valves
Part 3 Vessels
Part 4 Tanks
Part 5 Pumps
Part 6 Compressors
Part 7 Heat Exchanger
Part 8 Reactors
Part 9 Columns (Towers)
Part 10 Flares
Part 11 Electrical Equipment
Part 12 Cooling Tower
Part 13 Utility Systems
Part 14 Human Factors
Part 15 Global Events

Vessel

- What if vessel pressure increases?
- What if vessel pressure decreases?
- What if vessel level increases?
- What if vessel level decreases?
- What if vessel LAH fails?
- What if vessel LAL fails?
- What if vessel PAH fails?
- What if vessel PAL fails?
- What if vessel TAH fails?
- What if vessel TAL fails?

PART 1 PIPING

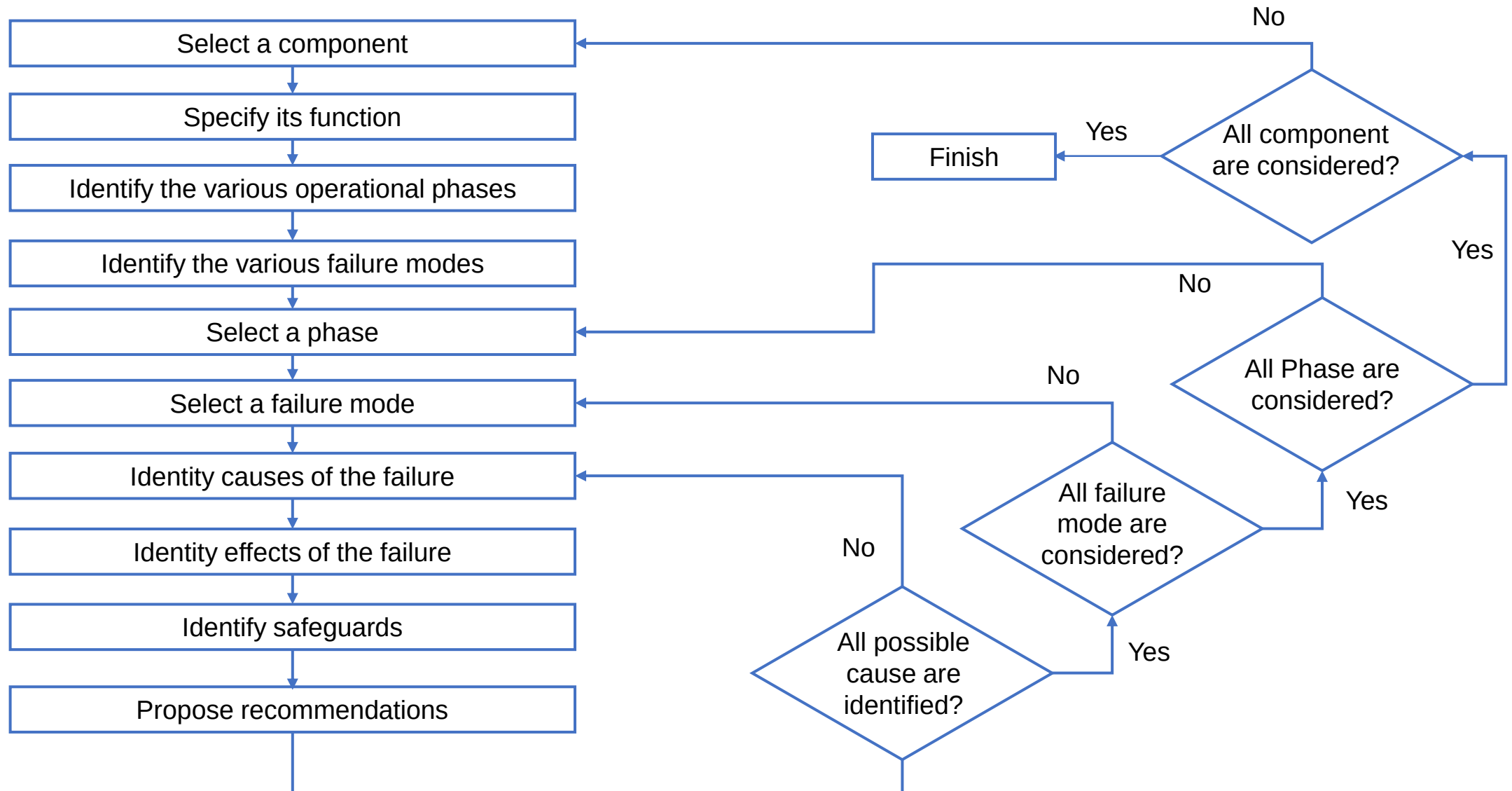
What-If checklist

- What if piping leaks?
- What if high pressure flammable, corrosive, or toxic gases leak into a liquid pipeline?
- What if piping is fractured?
- What if piping plugs?
- What if piping becomes fouled?
- What if moisture remains in piping?

- The What-if/Checklist is a combination of two methods 'What-if and Checklist' into one.
- A brainstorming approach in which a group of experienced people familiar with the subject process ask questions or voice concerns about
- The method is similar to What-if alone, with the difference being that broad categories of types of concerns are used to structure the analysis
- Study can begin with a checklist to which 'What-if' questions are applied to round out gaps in the list.

Failure Mode and Effects Analysis (FMEA)

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Failure Mode and Effects Analysis (FMEA)

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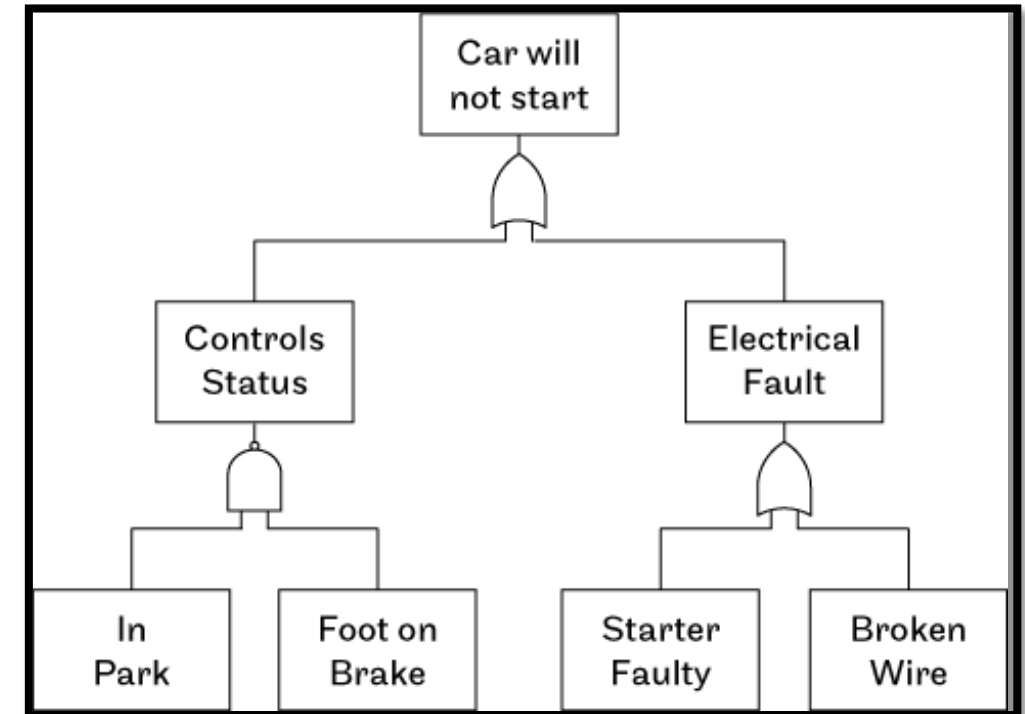
Applying Wax – Process FMEA

Process Function	Potential Failure Mode	Potential Effect(s) of Failure	Severity	Criticality	Potential Cause(s)/ Mechanism(s) of Failure	Occurrence	Current Process Controls	Detect	RPN	Recommended Action(s)	Responsibility & Target Completion Date	Action Results			
												Actions Taken	Score	Occurrence	Damage
Manual application of wax inside door To cover inner door, lower surfaces at minimum wax thickness to retard corrosion	Insufficient wax coverage over specified surface	Deteriorated life of door leading to: 1) Unsatisfactory appearance due to rust through paint over time, 2) Impaired function of interior door hardware	7		Manually inserted spray head not inserted far enough	8	Visual check each hour-1/shift for film thickness (depth meter) and coverage	5	280	Add positive depth stop to sprayer	Mfg Engineering 10/15/98	Stop added, sprayer checked on line	7	2	5
										Automate spraying	Mfg Engineering 12/15/98	Rejected due to complexity of different doors on same line			
					Spray heads clogged due to: 1) Viscosity too high, 2) Temperature too low, 3) Pressure too low	5	Test spray pattern at start-up and after idle periods, and preventive maintenance program to clean heads	3	105	Use Design of Experiments (DOE) on viscosity vs. temperature vs. pressure	Mfg Engineering 10/01/98	Temperature and pressure limits were determined and limit controls have been installed - control charts show process is in control Cpk=1.85	7	1	3
					Spray head deformed due to impact	2	Preventive maintenance programs to maintain head	2	28	None					
					Spray time insufficient	8	Operator instructions and lot sampling (10 doors per shift) to check for coverage of critical areas	7	392	Install spray timer	Maintenance 09/15/98	Automatic spray timer installed - operator starts spray, timer controls shut-off - control charts show process is in control Cpk=2.05	7	1	7

Fault –Tree Analysis

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- FTA draws a picture (model) that shows what undesirable outcomes might result from a specific initiating event (for example, a pipe rupture in a pipe rack)
- It uses graphics and symbols to show the possible order of events which might result in an accident.
- This method is sometimes used in accident investigations to determine probable causes.



Hazard Evaluation Techniques

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Project Phase/Method	Checklist	What-If	HAZOP	FMEA
R&D		✓		
Conceptual	✓	✓		
Pilot plant operation	✓	✓	✓	✓
Detail engineer	✓	✓	✓	
Construction/start-up	✓	✓		
Routine operation	✓	✓	✓	✓
Expansion or Modification	✓	✓	✓	✓
Incident investigation		✓	✓	✓
Decommissioning	✓	✓		

✓ Commonly used

Hazard Evaluation Techniques

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HAZOP	What-If/Checklist	FMEA	FTA	ETA
By deviation	By checklist item	By component	By loss event	By cause
<i>Best for process operations</i>	<i>Best for relatively standard operations</i>	<i>Best for mechanical and electrical systems</i>	<i>Best for complex systems/operations</i>	<i>Best to study one or only a few causes</i>
Good for continuous and procedure-based operations	Good for continuous and procedure-based operations	Good for continuous operations	Good for continuous operations; possible for procedure-based	Good to analyze administrative and engineering controls
Higher level of effort	Lower level of effort	Higher level of effort	Highest level of effort	Higher level of effort
<i>Can analyze complex processes with multiple safeguards</i>	<i>Mostly appropriate for simpler operations</i>	<i>Best analyzes processes with single-point failures</i>	<i>Can analyze complex processes with multiple safeguards</i>	<i>Can analyze complex processes with multiple safeguards</i>
Distinguishes between causes and safeguards	Does <u>not</u> distinguish between causes and safeguards	Does <u>not</u> distinguish between causes and safeguards	Distinguishes between causes and safeguards	Distinguishes between causes and safeguards
Only looks at causes that could lead to deviations	Only studies causes from checklist and what-if questioning	Looks at all failure modes of all components	Only studies causes and safeguards related to top event	Looks at all safeguards protecting against cause

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End of Chapter 2 : Overview PHA techniques

Technical and Safety service (Q-TS-TS)

PTT Global Chemical PCL